



$$\int e^{2x} dx = \int \underline{\underline{e^t}} dx \quad ? \Rightarrow$$

$$2x = t$$

$$\underline{d(2x)} = dt$$

$$2 dx = dt$$

$$dx = \frac{dt}{2}$$

$$\int \frac{e^t dt}{2} = \frac{e^t}{2} = \frac{e^{2x}}{2} \quad \downarrow$$

$$\left(\frac{e^{2x}}{2} \right)' = \frac{e^{2x} \cdot 2}{2} = \underline{e^{2x}}$$

$$\int \sin 3x dx = \int \frac{\sin t}{3} dt =$$

$$- \frac{\cos t}{3} = - \frac{\cos 3x}{3}$$

$$\int \cos \frac{x}{2} dx = 2 \sin \frac{x}{2}$$

$$\int (\cos^2 x \sin x - \cos x \sin x) dx =$$

$$\int \sin x (\cos^2 x - \cos x) dx$$

$$t = \cos x$$

$$dt = d(\cos x) = -\sin x dx$$

$$dx = \frac{dt}{-\sin x}$$

$$\int \frac{\cancel{\sin x} \cdot (t^2 - t) dt}{\cancel{-\sin x}} =$$

$$\int (t - t^2) dt = \frac{t^2}{2} - \frac{t^3}{3}$$

$$t = \cos x$$

$$C + \frac{\cos^2 x}{2} - \frac{\cos^3 x}{3} = F(x)$$

$$\begin{aligned} F'(x) &= \frac{2\cos x}{2} \cdot (-\sin x) - \frac{3\cos^2 x}{3} \cdot (-\sin x) \\ &= -\sin x \cos x + \cos^2 x \sin x \end{aligned}$$

$$\int (\sin^3 x \cdot \cos x - \sin^2 x \cos x + \sin x \cos x) dx$$

$$F(x) = \frac{\sin^4 x}{4} - \frac{\sin^3 x}{3} + \frac{\sin^2 x}{2} + C_1$$

$$\int \frac{\ln x}{x} dx$$

$$\ln x = t$$

$$d \ln x = dt$$

$$\frac{1}{x} dx = dt$$

$$dx = x dt$$

$$\int \frac{t}{x} \cdot x dt = \frac{t^2}{2} + C$$

$$\int \frac{dx}{1 + \sqrt{x+1}}$$

$$t = \sqrt{x+1}$$

$$dt = d\sqrt{x+1} =$$

$$\frac{dx}{2\sqrt{x+1}} \Rightarrow$$

$$dt = \frac{dx}{2t}$$

$$dx = 2t dt$$

$$\int \frac{2t}{1+t} dt \quad \textcircled{=}$$

$\textcircled{=}$

$$\frac{2t}{1+t} = \frac{2t+2-2}{1+t} =$$

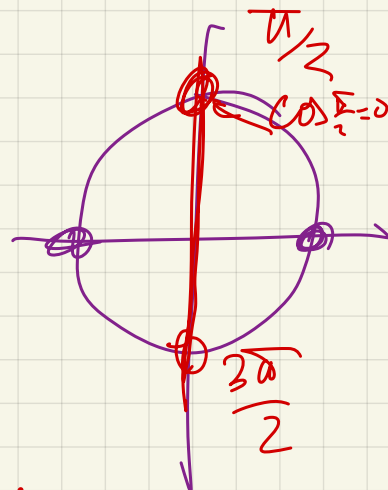
$$\frac{2(t+1)-2}{1+t} = \frac{2(\cancel{t+1})}{\cancel{t+1}} - \frac{2}{t+1}$$

$$= \left(2 - \frac{2}{t+1} \right) dt$$

$$\int 2 dt - \frac{2 dt}{t+1} =$$

$$u = t+1$$

$$du = dt$$



$$\int 2 dt - \int \frac{2 du}{u} =$$

$$= 2t - \frac{2u^2}{-2} = 2t + u^2 =$$

$$2\sqrt{x+1} + \frac{1}{\sqrt{x+1} + 1} + C$$

$$\int \frac{\sqrt{x}}{1+\sqrt{x}} dx$$

$$d\sqrt{x} = dt$$

$$\frac{1}{2\sqrt{x}} dx = dt$$

$$f(x) = x$$

$$\sqrt{x} = t$$

$$\left(\sqrt{f(x)} \right)' = \frac{1}{2\sqrt{f(x)}} (f(x))'$$

$$\frac{dx}{2t} = dt$$

$$dx = 2t dt$$

$$d\sqrt{f(x)} = \frac{1}{2\sqrt{f(x)}} df(x)$$

$$\int \frac{t \cdot 2t}{1+t} dt$$

$$= \int \frac{2t^2}{1+t} dt$$

$$\frac{t^2}{1+t} = \frac{t^2 - 1^2 + 1^2}{1+t} = \frac{(t-1)(t+1) + 1}{1+t}$$

$$2 \int \left(t-1 + \frac{1}{1+t} \right) dt =$$

$$2 \left(\int t dt - \int 1 dt + \int \frac{1}{1+t} dt \right)$$

$s = 1+t$
 $ds = dt$

$$\int \frac{1}{s} ds =$$

$$2 \frac{t^2}{2} - 2t + 2 \ln(s) = \ln s$$

$$= x - 2\sqrt{x} + 2 \ln(1 + \sqrt{x}) + C$$

$$1 - \frac{2}{2\sqrt{x}} \Rightarrow 2 \frac{1}{1+\sqrt{x}} \cdot \left(\frac{1}{2\sqrt{x}} \right)$$

$$1 - \frac{1}{\sqrt{x}} + \frac{1}{\sqrt{x}(1+\sqrt{x})} =$$

$$\frac{\sqrt{x}-1}{\sqrt{x}} + \frac{1}{\sqrt{x}(1+\sqrt{x})} =$$

$$\frac{(\sqrt{x}-1)(1+\sqrt{x})+1}{\sqrt{x}(1+\sqrt{x})} = \frac{x-\cancel{1}+\cancel{1}}{\sqrt{x}(1+\sqrt{x})}$$

$$= \frac{\sqrt{x}}{1+\sqrt{x}}$$

$$\int x^2 \sqrt{x^3+1} dx$$

$$\begin{aligned} \sqrt{x^3+1} &= u \\ d(\sqrt{x^3+1}) &= du \\ \frac{1}{2\sqrt{x^3+1}} \cdot 3x^2 dx &= du \\ \sqrt{x^3+1} &= t \end{aligned}$$

$$\begin{aligned} x^3+1 &= u \\ dx &= \frac{du}{3x^2} \end{aligned}$$

$$\int x^2 \sqrt{u} \frac{du}{3x^2} =$$

$$\int \frac{\sqrt{u}}{3} du$$

$$\sqrt{u} = u^{1/2}$$

$$= \frac{u^{3/2}}{\frac{3}{2} \cdot 3} = \frac{2 \cdot u^{3/2}}{9} =$$

$$\frac{2 \sqrt{(x^3+1)^3}}{9} + C$$

$$\int \frac{dx}{\sqrt{x^2+4x+5}}$$

$$\sqrt{x+2x^2} \neq \sqrt{x} + \sqrt{2x^2}$$

$$\boxed{x^2+4x+4} + 1 = (x+2)^2 + 1$$

$$\int \frac{dx}{\sqrt{(x+2)^2+1}}$$

$$t = x+2 \\ dt = dx$$

$$\int \frac{dt}{\sqrt{t^2+1}}$$

$$t = \operatorname{th} u \\ dt = d \operatorname{th} u =$$

$$\boxed{\frac{1}{\tan^2 u + 1} = \cos^2 u}$$

$$\frac{1}{\cos^2 u} du$$

$$\int \frac{\cos u \cdot du}{\cos^2 u} = \int \frac{du}{\cos u}$$

$$\left(\frac{\sin x}{\cos x} \right)' = \frac{\cos x \cdot \cos x - (-\sin x) \cdot \sin x}{\cos^2 x} =$$

$$\frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$\int \frac{du}{\cos u}$

Summe?

Kame gegen a journey

$$\int \frac{dt}{\sqrt{t^2+1}} = \ln(t + \sqrt{t^2+1})$$

$$= \ln(x+2 + \sqrt{(x+2)^2+1}) + C$$